COMPUTING & DISCRETE STRUCTURES III / INTEGRATIVE TASK 1

ResumeLens

From a resume to traceable qualification patterns.

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4 FSTs | 4 profile automata
26 canonical symbols / 56 aliases

TEXT
UTF-8 resume
Original wording

▸ REGEX
Extract
Exact evidence

▸ FST
Normalize
Canonical skills

▸ AUTOMATA
Recognize
Four profile patterns

▸ TEXTX
Validate + render
DSL to candidate report

01 / TEXT TO EVIDENCE

What does the text actually say?

Question: can explicit formal models handle varied wording while preserving the evidence behind every result?

```
Technical Skills:
JS, React.js, NodeJS, Postgres, Git.
```

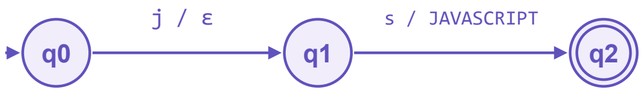
```
raw: "JS"   start: 238   end: 240
```

Positions use the LF Alex Sample reference.
Positive mentions and exclusions stay separate.

02 / EQUIVALENT WORDING

One alias, one canonical symbol

M = (Q, Σ, Γ, δ, ω, q0, F)




```
JS -> js -> JAVASCRIPT
```

Schematic path: consume the entire alias; only the last edge emits a canonical token. Empty output still consumes an input character.

Q: alias-path states Σ: casefolded characters
Γ: canonical symbols δ: character transitions
ω: edge outputs q0: start F: full-alias ends

Own: Type Script -> TYPESCRIPT | Git SCM -> GIT
Structured Query Language -> SQL. Full models: docs/models/

03 / CANONICAL SYMBOLS

When does a profile accept?

A = (Q, Σ, δ, q0, F) Type: ε-NFA

Full Stack
JS/TS + frontend + backend + database + Git

Machine Learning
Python + Pandas/NumPy + library + relational DB + Git
Library: Scikit-learn OR TensorFlow OR PyTorch

Backend
Compatible pair + database + REST API + Git

Data Engineer
Python + database + ETL/Airflow/Spark + Git

Q: states Σ: canonical skills δ: includes ε edges
q0: initial state F: accepting states

Bucket order stabilizes input. Compatible Backend branches constrain pairs. Several profiles may accept.

04 / EBNF AND COMPACT DSL EXAMPLE

Valid syntax before visualization

```
candidate {
  name "Alex Sample";
  contacts {} education {} experience {}
  skills { skill JAVASCRIPT; skill REACT;
    skill NODE_JS; skill POSTGRESQL; skill GIT; }
  accepted_profiles { profile FULL_STACK_DEVELOPER; }
}
```

```
Record = "record", JSONString, ";" ;
Education = "education", "{", {Record}, "}" ;
```

Keywords/tokens: terminals. Record/Education: nonterminals.
textX + semantic checks -> validated DSL -> escaped HTML.

05 / SEPARATE EVALUATIONS

Measured, with visible limits

14-case workflow regression **14/14**

Candidate data, spans, links, profiles and exports.

18-case preannotated evaluation **14/18**

14 exact cases; 4 observed disagreements retained.

Mentions: TP 94 / FP 3 / FN 3
Profiles: TP 17 / FP 1 / FN 2

23 historical extraction cases: 15 + 6 + 2.

VERIFICATION SNAPSHOT

Executable evidence

191 tests / 190 pass / 1 skip

Windows regression; source spans and files checked.

Bounded retries handle transient Windows file locks.

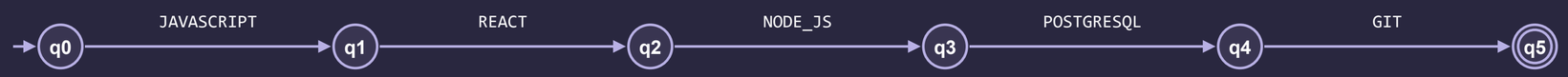
INTERPRETATION / NEXT STEPS

What the models cannot prove

- **Rust:** Unknown technology outside the catalog.
- **Double negation:** A positive Python claim can be missed.
- **Source attribution:** Job requirements can be mistaken for skills.
- **Pandas wording:** Bounded context misses some technical prose.

Next: independent real data and profile-domain validation.

ACCEPTING PATH
Schematic profile pattern



FULL STACK ACCEPTED

REPRODUCIBILITY Complete machines: docs/models/ and docs/automata/ | EBNF: docs/candidate-language.md | Evaluation: data/

REFERENCES Mohri (1997), Computational Linguistics 23(2), 269-311. Yu, Guan & Zhou (2005), ACL, 499-506. doi:10.3115/1219840.1219902

Synthetic regression, not population accuracy. Structure does not verify claims. No ranking or hiring decision is inferred.